

Expanded-Window Anchored Zonotope Obstruction

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Abstract

RH-254 doubles the resolved candidate margin and supplies 16 new roots at each archived endpoint. We now perform the required anchored reachability audit rather than a zero-target fit. After discarding incomplete boundary pairs, every conjugate shell receives a single-use weight $0 \leq w_j \leq 1$. The resulting convex zonotope has zero passes at all 32 endpoints. Its weighted distance from the deterministic target ranges from 0.143585 to 0.423998, with primal–dual gaps below 5.83×10^{-15} . Hence every prefix and each of the 62,030,604,700 eligible binary subsets also fails. This is a finite margin-32 obstruction, not a global nonexistence theorem.

1 Anchored expanded class

For orders $n = 2, \dots, 12$, let $a = (a_n)$ be the deterministic target of RH-243 [1]. At a fixed noise/channel endpoint, remove the Perron and parity traces from the full matrix powers and write the remaining base vector as b . If S_j is a conjugate-complete shell from RH-254, put

$$v_j = \left(\sum_{\mu \in S_j} \mu^n \right)_{n=2}^{12}. \quad (1)$$

All v_j are real up to the archived conjugacy tolerance.

The expanded single-use class is the zonotope

$$\mathcal{Z} = \left\{ b - \sum_j w_j v_j : 0 \leq w_j \leq 1 \right\}. \quad (2)$$

Its anchored distance is measured by

$$d(\mathcal{Z}, a) = \min_{0 \leq w \leq 1} \sum_{n=2}^{12} \frac{|b_n - a_n - (Vw)_n|}{n}. \quad (3)$$

Proposition 1.1 (Box dominance). *Every shell prefix, contiguous shell interval, and binary single-use subset is contained in the box class (2). Therefore $d(\mathcal{Z}, a) > \varepsilon$ excludes all such selectors at tolerance ε .*

Proof. Each listed discrete selector has a weight vector with coordinates in $\{0, 1\}$, hence lies in $[0, 1]^J$. Minimizing over the larger box can only decrease the distance. $\square$

2 Primal–dual audit

The weighted ℓ^1 problem (3) is a linear program. We solve both its box primal and explicit zonotope dual, then compare their objectives. The shell-complete ranks range from 33 to 64; no incomplete conjugate fragment enters the optimization [3].

quantity	value
endpoints	32
eligible prefixes	836
eligible binary subsets	62,030,604,700
prefix passes	0
box passes	0
box distance range	0.143585–0.423998
distance/tolerance range	10.17–117.66
minimum failure margin	0.142335
maximum primal–dual gap	5.83×10^{-15}

All 32 expanded distances improve relative to the frozen RH-248 box [2]; the improvement ranges from 2.43×10^{-7} to 0.00583514. Improvement is not reachability: the best expanded point remains more than ten local tolerances away.

Corollary 2.1 (Scoped expanded-window obstruction). *At each archived endpoint, no prefix or binary subset of the margin-32 conjugate-complete shells reaches the deterministic anchor within the local noise tolerance.*

Proof. The computed box distance is larger than tolerance at all endpoints, and Proposition 1.1 applies. The dual objectives agree with the primal values to the stated floating tolerance. $\square$

3 Boundary and next route

The conclusion is confined to the finite margin-32 single-use shell class. It does not exclude a still larger resolved window, unbounded weights, or a signed/complex selector forced by invariant quotient algebra. It also does not provide interval enclosures for the eigensolver roots.

The live next route is therefore structural rather than another box scan: derive signed or complex coefficients from a quotient identity and audit whether they form a legal selector. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, or RH implication is claimed.

References

- [1] Bin Wang. A deterministic numerator coefficient-anchor dictionary. *Preprint*, 2026. RH-243 preprint.
- [2] Bin Wang. Anchored shell-zonotope reachability obstruction. *Preprint*, 2026. RH-248 preprint.
- [3] Bin Wang. Expanded resolved candidate-window atlas. *Preprint*, 2026. RH-254 preprint.